

Quarterback Spending and NFL Team Success (2013–2025)

This report examines the relationship between quarterback (QB) salary allocation and team success in the National Football League using team-level data from the 2013 through 2025 seasons. Team success is evaluated using regular-season win percentage and playoff appearances. The goal of the analysis is to assess whether higher QB spending improves outcomes and how those effects differ across relative cost environments.

Data and Methodology

The dataset consists of one CSV file per NFL season from 2013 through 2025, each containing team-level salary allocations by position. All seasonal files were programmatically combined into a single dataset prior to analysis. Quarterback salary values were cleaned and converted to numeric format. Regular-season win percentage was taken directly from the dataset to account for seasons with different numbers of games, and playoff appearance was converted into a binary indicator.

Linear regression was used to model the relationship between quarterback spending and win percentage. Because QB salaries are highly skewed, spending was log-transformed prior to modeling. Logistic regression was used to estimate the relationship between QB spending and the probability of making the playoffs. To evaluate cost efficiency, teams were classified as low-cost QB teams if their QB spending fell in the bottom third of the dataset.

Results

Across all teams, higher quarterback spending is associated with a modest increase in regular-season win percentage. The linear regression model estimates a positive relationship between log-transformed QB spending and win percentage, though the model explains only a small portion of overall performance variation ($R^2 \approx 0.014$).

Logistic regression results indicate that increased QB spending improves playoff odds. The estimated odds ratio of approximately 1.29 suggests that a proportional increase in quarterback spending is associated with a meaningful increase in the likelihood of reaching the postseason.

When teams are grouped by relative QB cost, high-cost QB teams reached the playoffs more frequently on average (approximately 43%) than low-cost QB teams (approximately 34%). A logistic regression using only QB cost classification finds that low-cost QB teams have roughly 30 percent lower odds of making the playoffs than high-cost QB teams.

Conclusion

The findings indicate that elite quarterbacks, who typically command higher salaries, raise baseline playoff probability. However, quarterback spending alone explains only a small share of team performance, underscoring the importance of overall roster construction. These results highlight the trade-offs teams face between investing heavily in quarterback talent and maintaining spending

flexibility across the rest of the roster.